Fulbright University Vietnam

COMPUTATIONAL SOCIAL MEDIA

Final Project:

Impact of TikTok on ADHD Symptoms

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Abstract

As TikTok becomes increasingly prevalent and even addictive among regular users, understanding the impact of TikTok usage on mental health is necessary to cultivate healthy and responsible user practices. Previous studies in psychology and neuroscience have shown that there are strong correlations between social media, and in particular TikTok usage and psychological distress in Vietnam. However, there has been little statistical research focusing on the correlations between Attention Deficit Hyperactivity Disorder (ADHD) and TikTok usage. To fill this gap, our study uses a dataset of 201 responses to an online survey asking participants for their time spent using TikTok, usage behaviors (interaction), and self-reported symptoms of ADHD. This research aims to investigate the correlation between the amount of time and usage behaviors on the attention span of TikTok users. Our findings demonstrated that daily TikTok usage is significantly linked to increased attention difficulties. We further analyzed how the period of the day affects attention span, and the results showed that users who reportedly use TikTok from 4 P.M. to 9 P.M. report more attention difficulties than those who usually use TikTok in the morning, from 6 A.M. to 9 A.M. Frequency of interaction while using TikTok, however, showed no significant link with attention span or impulsivity. These findings reaffirm previous studies that longer time using TikTok is associated with lower attention span and impulse control.

Keyword

ADHD: Attention Deficit Hyperactivity Disorder is a mental disorder affecting mainly children. Symptoms of ADHD include inattention (not being able to keep focus), hyperactivity (excess movement that is not fitting to the setting) and impulsivity (hasty acts that occur in the moment without thought) [4].

1 Introduction

TikTok enjoys such huge popularity and a large number of frequent users. With short-form videos (from 15 to 60 seconds) as its signature, TikTok has become a platform for innovative content such as dance challenges, cooking tutorials, or short comedy. It offers algorithms that recommend videos that match with users' likings to maintain high user engagement, and displays videos from worldwide trends that give people a sense of global popular culture [1].

By 2020, it is reported that TikTok had approximately 800 million active users around the world, and this number continued to grow enormously [12]. As its debut in the market coincided with the COVID-19 pandemic period, many users regarded TikTok as a "feel-good space" where they had "access to relatable content" that spoke to their "escapist desires and needs" [11]. For daily use, many studies have pointed out that TikTok has an addictive nature owing to the algorithm-generated cycle of videos. The "For You Page" of TikTok offers streams of videos in a continuous cycle of videos that appeal to users' preferences [10]. Users who often search for baking videos, like baking tutorial videos, or search for names of desserts would find countless videos about baking on their TikTok For You Page.

Although TikTok establishes strong connections to users by recommending a seemingly never-ending cycle of videos serving their peculiar tastes, the addictive nature of TikTok has raised concerns. The health risks of TikTok addiction has been a topic of research interest. The National Institute of Health pointed out that the effects of TikTok had an impact on the developing minds of younger users. Research on TikTok usage among high school students revealed that TikTok negatively impacts the attention span of students. The dopamine stimulated by using TikTok is high, hence students will feel dependent on such a source of comfort and tempted to use TikTok even during class time [10]. Neurological research suggests that the consumption of TikTok stimulated content affects the brain's capacity to process and store information [10].

Ever since its official launch in Vietnam in 2019, TikTok has attracted a massive number of users. Research about the impact of social media on mental health in Vietnam mainly focuses on disorders like stress, fear of missing out caused by online content, and addiction, or comparing the addiction levels between different platforms. According to psychological research using Kessler

Psychological Distress Scale in Vietnam, there are strong correlations between social media usage and psychological distress [3]. TikTok and Instagram are reportedly more harmful to students' mental health than Facebook and Zalo [13]. Social media addiction scores are higher among those who feel fear of missing out when neglected by online peers [3].

However, there has been less research focusing on the impact of TikTok and attention span among Vietnamese users. To cover the less examined areas of research, this paper aims to answer the following research question:

RQ: How do the frequency of interaction and the amount of time spent on TikTok correlate with ADHD symptoms, specifically attention span, among the young generation?

Contribution

We conducted a study with a dataset of 201 participants in Vietnam with the goal of identifying the total duration they spend using TikTok every day, their behaviors (interactions like comment or repost) when using TikTok, and ADHD self-reported symptoms. To be able to measure accurately, we used the TikTok Addiction Scale (TTAS) - a self-reported behavioural scale based on TikTok use, to create a set of questions and answers based on how the research "TikTok Addiction Scale: Development and Validation" is used to measure TikTok addiction through a validated measurement. This TTAS scale uses a Likert scale from 1 to 5, with the highest index equivalent to reduced concentration and high TikTok addiction [5]. We first conducted a descriptive analysis that demonstrated the time usage of Tiktok's users and then examined the distribution of the user's daily usage duration with the attention deficit scale to get an overview of the dataset. Further, we statistically analyzed usage time by mapping it into a time frame from low to high use for better understanding the impact of time on attention span, then examined the relationship between time usage and cognitive score through linear regression. Moreover, we analyzed the impact of period usage during the daytime and found that the evening was the time when users had lower concentration. Finally, we found out that the users' interaction frequency does not bring up the effect of worsened attention span as much as it tends to for users' impact, which suggests that users are less likely to perceive the problem of concentration and are unaware of the need to reduce the time spent on the platform.

2 Background and Related Work

The growing popularity of TikTok and other short-form video (SFV) platforms among young users has raised concerns about their impact on consumers' mental health. As such, recent research has also focused primarily on general psychological issues such as anxiety and depression associated with TikTok use. For example, the systematic reviews (Sha et al., 2023) found consistent associations between problematic TikTok use and higher levels of emotional distress. Similarly, at previous studies investigating the association between TikTok use and attention span have focused primarily on passive consumption patterns, suggesting that heavy use is associated with reduced attention span in adolescents. However, relatively few studies have explored how TikTok use relates to specific cognitive symptoms, particularly those associated with Attention Deficit/Hyperactivity Disorder (ADHD).

The relationship between TikTok interaction patterns and ADHD symptoms is important because this age group is developmentally vulnerable to attentional disruptions. In Kim et al.'s (2021) study, it conducted an extensive literature review and found a consistent pattern linking high-frequency internet use to reduced attentional performance. While the study did not focus specifically on TikTok, it highlights how fast-paced digital content can overstimulate users and shorten their attention spans over time. This is because users become conditioned to seek constant novelty and instant gratification, leading to a weakening of their tolerance for longer, less stimulating tasks. Such behavioral conditioning closely mirrors the distraction and inattention observed in ADHD symptoms, suggesting a potential cognitive pathway by which SFV use may impact mental health.

Further evidence comes from research on user engagement and addiction mechanisms. Li and Chen (2022) explored TikTok addiction using a system quality perspective, with features such as personalized content and algorithm-driven entertainment being strongly associated with compulsive use. Their findings differentiated between passive consumption such as simply watching videos and active engagement such as liking, commenting, reposting, or creating content. Active users were more likely to experience emotional dependence and stronger psychological immersion, which in turn was associated with reduced attention span. These findings suggest that engagement intensity is not just about time spent but also is important role in cognitive outcomes and supporting the hypothesis that active engagement in TikTok may exacerbate ADHD-related symptoms.

Collectively, these studies reveal a gap in the specific role of user engagement and content creation behaviors that remains underexplored. This gap involves additional cognitive load for users, and it potentially exacerbating attention-related symptoms. Recognizing this gap, the current study aimed to investigate how frequency of engagement and the amount of time spent on TikTok correlate with ADHD symptoms in adolescents, with particular attention to differences between passive and active engagement patterns. By focusing on metrics of active engagement rather than just passive use, this project provides a more nuanced understanding of how different patterns of TikTok use may contribute to cognitive outcomes. In doing so, the project addresses an underexplored aspect of digital media research and aims to provide more actionable insights for mental health interventions targeting younger generations.

In Vietnam, the growing popularity of short-form videos (SFVs) has raised issues related to user addictions, especially in younger adults. In the essay by Nguyen (2023) [7], the author analyzes SFVs addiction using the stress-coping theory, endorsing a new psychological approach to the issue of digital addiction. The study proposes a comprehensive model that includes stressors: coping mechanisms of social isolation and anxiety; escapism and social interaction; and outcomes like perceived enjoyment and mood regulation. In SEM analysis conducted on a Vietnamese sample, the results indicated that perceived enjoyment and mood regulation directly heightened SFV addiction, while escapism and social interaction were indirect influences. A key observation is the insignificant effect of neuroticism, contrary to much prior literature on digital addiction. The study is methodologically robust and adds value to research on behavioral addiction issues, though the use of self-reported data and ethnocentric bias are weaknesses in scope. Further research could build on such findings by investigating the role of culture with more robust longitudinal frameworks.

3 Data Collection

The data for this study were collected from an original online survey created to measure TikTok use and ADHD symptoms in young Vietnamese adults. To conduct the survey, we used the Vietnamese language and engaged with TikTok users aged 13 years and over. The questionnaire was composed of demographic questions (age, gender, occupation) and a few behavioral questions about the use patterns of TikTok and its psychological consequences.

The questionnaire has both multiple-choice and Likert scale questions, which allow quantification of attention and impulsivity-related experiences marked as subjective. The report questionnaire is distributed online (on social media and university networks) to make it a more diverse sample of participants who complete the questionnaire. The final dataset comprised 201 responses and, as a result, enables the use for subsequent analysis.

3.1 Data processing

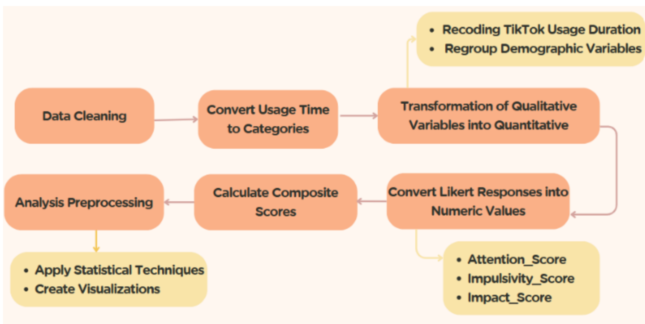

Figure 1: The overview data analysis of the study

Several methods were employed in the data cleanup to make the data more conducive for analysis, consistent, and understandable for statistical evaluation such as:

- **Standardization of Variables:** Survey item variable names were standardised to improve consistency and clarity in our data. For example, the question "Bạn trong độ tuổi nào?" was renamed "age_group", which made data management and analysis easier.
- **Categorical Encoding:** The daily_duration to name a few categorical variable were then converted into numerical values so they can be analyzed quantitatively. For example, answers such as "More than 3 hours" were coded as a 4 and Likert responses were recoded so that 1 (Never) $\rightarrow$ 5 (Always). This encoding made possible to apply wide range of statistical methods.
- **Composite Scoring:** Related items were aggregated into three composite variables to simplify the analysis and focus on broader constructs:
  Attention_Score: This score measured attention difficulties, incorporating items such as focus_difficulty and scroll_unaware.
  Impulsivity_Score: This score assessed impulsive behaviors, including task_switching and phone-restlessness.
  Impact_Score: This score captured the perceived negative impact of TikTok usage and the desire to reduce usage.

These processing steps allowed for reliable quantitative methods, and the data were ready for either descriptive or inferential statistics.

3.2 Dataset overview and variables

To facilitate analysis, the original survey questions were standardized and renamed as follows:

Original Survey Question (Vietnamese)	Renamed Variable
Bạn trong độ tuổi nào?	age_group
Giới tính	gender
Bạn đang là:	occupation
Thời lượng bạn sử dụng Tiktok 1 ngày là bao nhiêu?	daily_duration
Bạn thường sử dụng Tiktok vào thời điểm nào trong ngày?	time_usage
Bạn sử dụng mạng xã hội Tiktok chủ yếu để làm gì?	usage_purpose
Bạn có repost nội dung trên Tiktok không?	do_repost
Bạn có thường xuyên cuộn liên tục trên TikTok mà không nhận thức được thời gian trôi qua không?	scroll_unaware
Sau khi sử dụng TikTok trong thời gian dài, bạn có cảm thấy khó tập trung vào công việc hoặc học tập không?	focus_difficulty
Bạn có gặp khó khăn trong việc duy trì sự chú ý vào một công việc hoặc hoạt động trong thời gian dài không?	attention_difficulty
Bạn có hay chuyển từ việc này sang việc khác mà không hoàn thành công việc ban đầu không?	task_switching
Bạn có thường cảm thấy bồn chồn, không thể ngồi yên khi không có điện thoại không?	phone_restlessness
Bạn có cảm thấy khó kiểm soát thời gian sử dụng TikTok, ngay cả khi biết nó ảnh hưởng đến công việc/học tập không?	control_usage
Bạn có nghĩ rằng TikTok ảnh hưởng tiêu cực đến khả năng tập trung của bạn không?	perceived_attention_impact
Nếu có cơ hội, bạn có muốn giảm thời gian sử dụng TikTok không?	Want_reduce_time

Figure 2: Survey questions to standardized variables

Demographic Variables:

- age_group: Age group categories (e.g., 13-17, 18-25, 25+)
- gender: Respondent's gender
- occupation: Participant's occupation

Usage Variables:

- daily_duration: The amount of time user spends on TikTok per day
- time_usage: The specific time (morning, mid-day, evening, mid-night)
- usage_purpose: Purpose of using TikTok
- do_repost: Whether user reposts

4 Method

Several important methodological rigor steps were included in the analysis:
Exploratory Analysis: Descriptive statistics

and visualizations are used to understand the demographics of survey respondents. We included the use of histograms or boxplots to better understand the distributions of Attention Score and Impulsivity Score responses.
Composite Score Calculation: The Attention, Impulsivity, and Impact Scores were derived from averaging a set of related questions. This parsimonious approach simplified the model to only focus on general constructs, which made the results potentially easier to interpret.
Statistical Testing:

- **The Kruskal-Wallis Test:** used to perform group comparisons by category variables (low, medium, and high user groups).
- **Spearman Correlation:** used to examine the monotonic relationship between TikTok usage time and cognitive metrics.
- **ANOVA test:** used to compare differences in cognitive scores between usage time groups and determine significant statistical differences.
- **Linear Regression:** used to model the predictive relationship between daily usage time and Attention Score.

5 Results

5.1 Overview of users and TikTok usage

1. Grouping users by time spent using TikTok
To get an overview of the participants' TikTok usage habits, we analyse the amount of time distribution of TikTok usage with answers from our dataset. Then, we visualized a bar chart (Figure 2) to represent the number of users by time usage: low - less than 1 hour, medium 1-3 hours, high - more than 3 hours. The results show that the number of users who watch Tiktok less than 1 hour/day is 45 users in the lowest group and the number of users who watch more than 3 hours/day is 65 users in the medium group, most of which are concentrated in the 1-3 hour/day time frame with about 95 users. This shows that the majority of users are in the medium usage group, indicating an average level of daily interaction with TikTok. This balanced distribution allows for meaningful comparisons between usage levels.

2. Relationship between usage time and cognitive scores

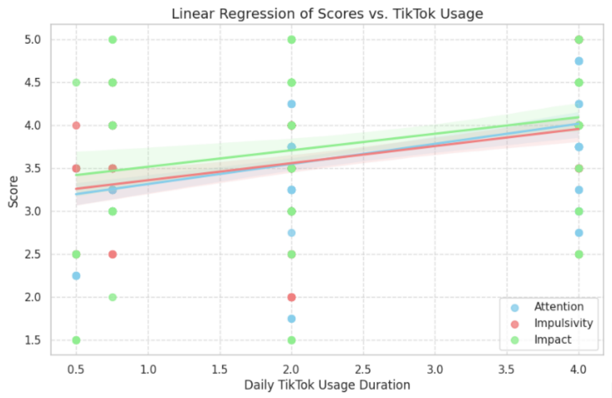

Figure 3: Linear regression scatter plot represents the comparison between cognitive scores (attention, impulsivity, impact) with TikTok time usage

From the characteristics of the number of users by duration, we continue to examine whether the factor between the time spent on Tiktok has a relationship with psychological behavior according to the behavioral measurement variables including attention - measuring the difficulty in concentrating (blue), impulsivity - measuring behavior (red), impact - perception of using Tiktok (green). To visualize the relationship, we illustrate the linear regression graph (Figure 3) to understand better how the amount of time spent on TikTok will affect user cognition. Furthermore, all three constructs show a positive linear relationship with the time spent on TikTok. This suggests that users who spend a lot of time on this platform will tend to have difficulty concentrating based on the positive linear relationship, increased attention bias, and increased impulsivity, but at a more moderate level.

5.2 Amount of time usage

1. Correlation between TikTok use and cognitive metrics
These trends are consistent with Spearman correlation and regression results, providing strong visual support for the hypothesis that

cognitive/behavioral difficulties increase with use.

Composite Variable	Spearman's ρ	p-value
attention_score	0.40	< 0.001
impulsivity_Score	0.36	< 0.001
impact_score	0.22	0.004

Table 1: The Spearman correlation result between daily time usage duration and cognitive score

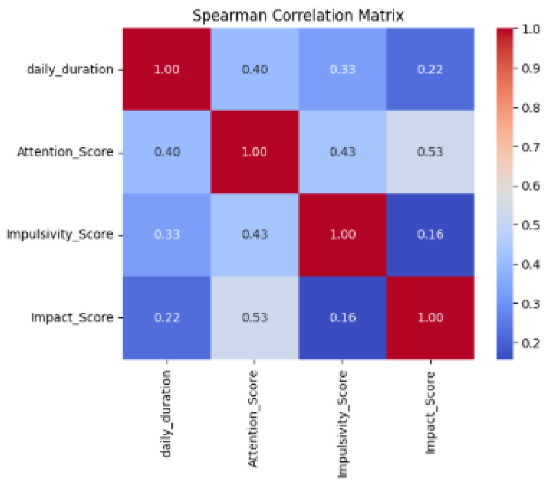

Figure 4: The Spearman correlation matrix

To further investigate the correlation between the time spent using TikTok and cognitive data, we conducted a spearman correlation analysis to examine the correlation between the variables. According to the results in Table 2 & Figure 4, it can be seen that the correlation matrix has a strong correlation between the attention score and the time spent using Tiktok (p $<$ 0.001, p = 0.40). (Note that the level of attention here refers to the level of attention loss, the higher the score based on the score, the greater the level of attention reduced according to the TTAS scale.) [5] Thus, it can be seen that when users tend to spend more time using TikTok, the difficulty of concentration also tends to increase, as well as the impulsive score also shows a positive correlation, while the impact score, reflecting cognition, has a weaker correlation (p = 0.004, p = 0.22). Through the matrix in Figure 4, the description is clearer, the trends are shown more clearly, the relationship between the concentration score and the impulsivity score shows that the impact of using TikTok for a long time will cause difficulty in controlling behavior, as well as through the impact points, we see that the user's level of restraint is still quite limited.

2. Differences in usage time by group in cognitive metrics

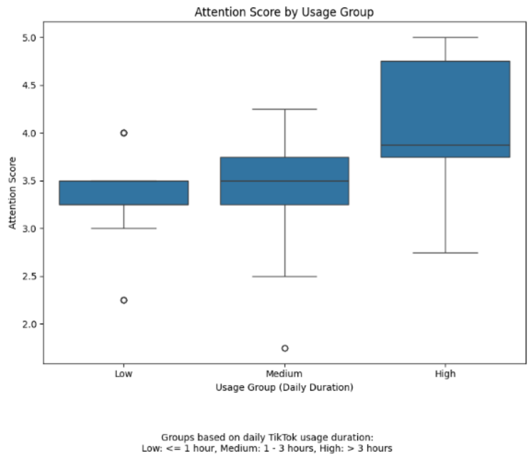

Figure 5: Differences in throughout daily TikTok time usage

To further explore differences across the entire time spent (Figure 5), we used the Kruskal-Wallis method to examine the magnitude of differences between groups of time spent on TikTok (low, medium, high). With the test results, we can see the difference between the groups, H = 31.01, along with the p-value $<$ 0.05 (p = 1.85e-07), indicating a statistically significant difference, indicating that usage time affects users' ability to concentrate. In addition, the box plot comparing the average scores of the attention index with the three groups of time spent shows more clearly and directly the magnitude of the effect. The trend of the average score tends to increase gradually with the usage time length from low usage (3.5) to high usage (4.5). However, in the group with high usage (more than 3 hours), we saw a clear difference in difficulty concentrating when using TikTok for long periods, as their IQR tended to have a wide range between those with high and low attention scores. Thereby, we found that although the increase in attention ability with time of use suggests that the longer the time of use, the higher the level of distraction, the finding of a wide IQR in the attention ability group suggests not only a linear relationship but also a clear difference in concentration difficulty between different levels of use.

3. Predicting the time using the linear regression model

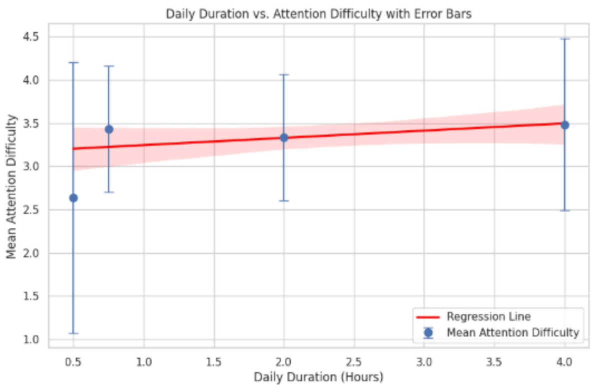

Figure 6: The linear regression model to determine daily TikTok usage time and attention score

The regression model uses the independent variable: time using TikTok (hours) & the dependent variable attention difficulty score, for the equation is:

Attention Score=3.0821+0.2341×(Daily TikTok Duration in hours)

The results of the regression equation show that the attention score increases by 0.2341 points for each hour of using TikTok with the regression line showing a positive relationship but there is no obvious increase. However, this model is still significant when the p-value <0.001 ($F = 48.76$) shows a positive level between time spent on TikTok and attention, showing that every hour increases, the attention span score will decrease. Although there are fluctuations around the data, the linear regression model demonstrates a correlation between the two variables, adding to the validity of the prediction of the correlation between the time spent on TikTok and the trends of decreased attention and behavior.

5.3 The differences in the distribution of cognitive daytime periods

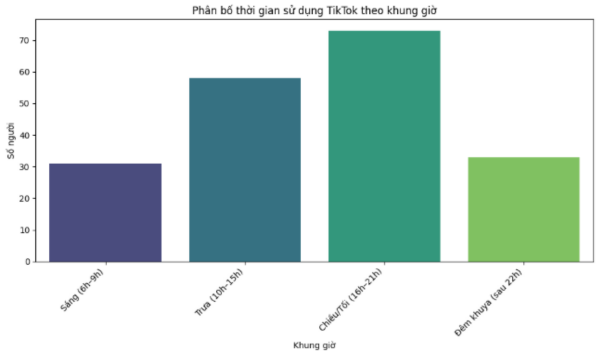

Figure 7: The number of TikTok users distributed by time frame

The bar chart in Figure 7 shows us the difference between each period, we can see that the highest usage occurs in the Evening (4–9 pm) with nearly 70 users, followed by Noon (10 am–3 pm). This is followed by morning (6–9 am) and late night (after 10 pm). This pattern reflects some of the other findings, from what we initially predicted that midnight would have the most users, the results showed that occasional users visit TikTok more in their free time after school and work.

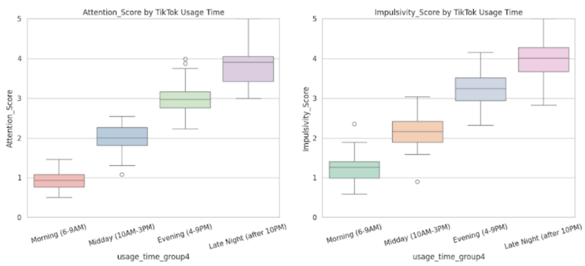

Figure 8: The comparison between attention score/impact score with TikTok usage time

Based on the previous finding, we wanted to dig deeper into how each attention point responds differently at each time of day. From Figure 8.1 & 8.2, the concentration scores of the sessions tend to increase over time from morning (average attention score at the lowest level of nearly 1) to late night, with the highest average score reaching almost 4. Furthermore, the dispersion - IQR of the afternoon and late night is more dispersed and unstable with a fairly widespread. Similar characteristics also appear in Figure 3.3, the tendency for the impulsive score to increase more than the concentration score in some periods, but still increase for the most part.

Both graphs show that users who use TikTok in the morning (6-9 am) have the highest level of concentration and better behavioral control than in other sessions, suggesting that as the evening progresses, TikTok users will be less focused and more impulsive. This shows similarities with users who tend to use TikTok more in the late afternoon and evening (in Figure 2), which suggests that increased time spent on TikTok may be responsible for the decreased ability to concentrate in the late afternoon and evening.

5.4 The influence of attention difficulty level by age groups

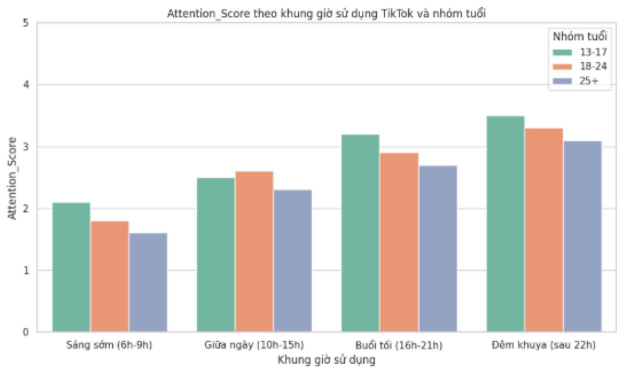

Figure 9: Correlation between concentration and time of use by age group

Through the time frames used in the above analysis, our team wanted to further analyze how age groups (13-17, 18-24, and 25+) would have differences in their use of TikTok. In the Image, the graph compares attention scores by time frame of TikTok use and age group. Note the key features, in all three age groups, the attention score is most affected in the evening (4-9 pm). Furthermore, although all three age groups tend to have a decrease in concentration level over time, the age group 13-17 tends to be most affected in terms of concentration level, as in the late afternoon, this group's score is always lower than the other groups. From the observation results, we can see that the evening time (4-9 pm) is the optimal time to use TikTok, especially for both young and old age groups, the 13-17 and 18-24 age groups tend to use TikTok more than the group after 25 years old. This represents the possibility that as the time spent on TikTok increases, the level of attention tends to decrease. This usage trend demonstrates that longer periods of use can harm sustained concentration, especially when periods of use extend into the evening or night.

5.5 The correlation between TikTok users' interaction frequency and cognitive score

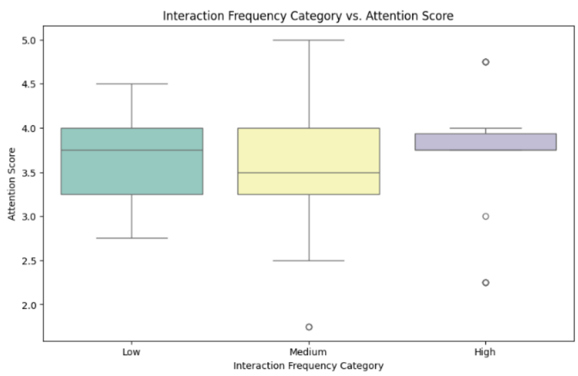

Figure 10: Comparison between low, medium, and high frequency of use with attention score

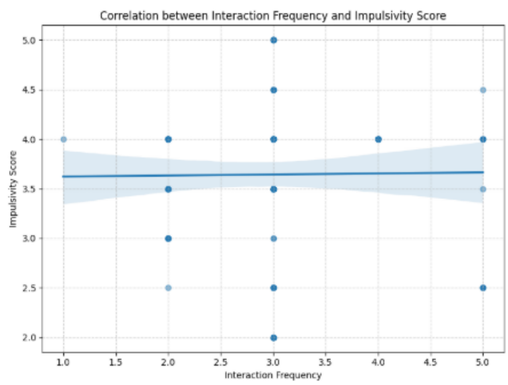

Figure 11: Comparison between the impulsivity score and interaction frequency

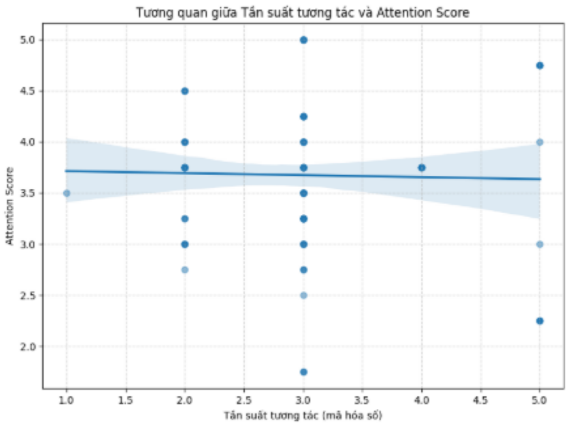

Figure 12: Comparison between the attention score and interaction frequency

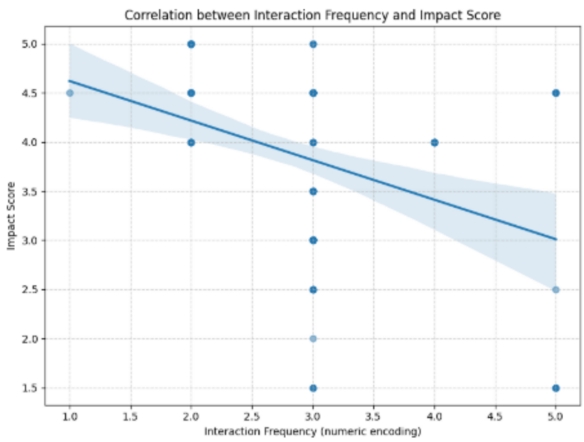

Figure 13: Comparison between the impact score and interaction frequency

The graph shows that the relationship between interaction frequency and psychological indicators in young users is not simply solidified by the assumption that more interaction is reduced in attention span. Specifically, in Figures 11 & 12, the impulsivity score and attention score are not significantly affected by interaction frequency. However, the impact score in Figure 13, which reflects the user's ability to feel extreme impact and desire to change behavior, tends to decrease as interaction frequency increases, suggesting that users who interact frequently may feel less impactful content, which may represent a reduced sensitivity to the content. In contrast, the ANOVA results (F = 17.52, p = 0.00) showed a statistically significant difference between the three frequency groups, as the low levels of interaction have fluctuations, while the medium and high groups maintained stable or higher levels of attention, noting that a reasonably compatible activity may support the maintenance of attention. Overall, the frequency of interactions with the platform did not show much of a relationship between frequency and attention difficulty, but there was a difference between actual performance and impact score.

6 Discussion

6.1 Interpretation of Key Findings

This analysis provides evidence for a statistically significant relationship between the frequency of interaction spent on TikTok and core symptoms associated with ADHD, especially attention difficulties and impulsive behavior.

- Attention_Score: $\rho = 0.40$, $p < 0.001$
- Impulsivity_Score: $\rho = 0.36$, $p < 0.001$
- Impact_Score: $\rho = 0.22$, $p = 0.004$

According to the linear regression model, each additional hour spent daily on TikTok was found to account for an increase of 0.2341 points in the Attention Score, with usage time accounting for 20% of the variance in attention difficulty ($R^2 = 0.20$). This affirms TikTok usage as a significant predictor of attention-related problems in youth.

Although the first Kruskal-Wallis test didn't show clear group differences ($p = 0.509$), another test looking at the time of day people use TikTok did show a big difference ($F = 13.033, p < 0.001$), which means that not just how much time, but how people use TikTok also matters for their attention.

6.2 Behavioral Interpretation and Usage Time

Users engaging with TikTok in the evening between 4 PM and 9 PM reported the highest Attention Scores, which could be due to increased emotional arousal during peak entertainment hours or a decrease in mental energy after a stressful day of studying and working. In contrast, people who use TikTok in the early morning from 6 AM to 9 AM noted lower Attention Scores, which could be attributed to less mental effort or distractions. In terms of frequency of interaction, users who actively engaged with TikTok (commenting, reposting, or sharing) showed high levels of impulsivity and attention problems. Active use of TikTok appears to increase cognitive load and shorten attention span more than passively scrolling through videos.

6.3 Comparison with Prior Research

The results of this study are consistent with previous research on the cognitive consequences of digital media, especially short, fast-paced content. Prior literature has shown that excessive media use and dopamine-triggering environments (like TikTok's recommendation algorithm) can disrupt attention regulation. Intensive usage patterns, such as frequent interaction, have also been linked to stronger emotional dependence and shorter attention spans, which further supports the findings of the current study. This research contributes to the existing literature by specifically linking both usage time and frequency of interactions on TikTok to measurable attentional outcomes, rather than focusing solely on social media in general or total screen.

6.4 Limitations

- The study's cross-sectional design prevents any conclusions about causality.
- The use of self-reported survey data may introduce recall bias or social bias.
- The Kruskal-Wallis test for usage levels was not significant in the initial analysis, possibly due to limitations in group size or categorization.

7 Conclusion

This research clearly shows a link between hours spent on TikTok, engagement with the platform, and expression of ADHD symptoms, more specifically attention-related issues among adolescents and young adults. The data directly supports the hypothesis that increased exposure to short-video content is associated with increased attention difficulties, suggesting that TikTok's interface and content format may contribute to attention difficulties.
These findings go beyond simple correlation:

- Active users (those who interact a lot) tend to have more difficulty with focus and self-control than those who just watch videos.
- There is a dose-response effect, meaning that more TikTok use is associated with more severe attention problems.
- Time-of-day also plays a crucial role, as people using TikTok in the evening tend to have higher attention difficulty than those using it in the morning.

Finally, these results show that not only the duration but also the type and timing of TikTok use have a significant impact. This highlights the need for more focus on digital well-being, especially for teenagers.

8 Future Implications

The findings of this study have several important implications for future research, education, and policies related to digital media use. First, extensive studies are needed to establish causal relationships and better understand the long-term effects of frequent TikTok use, particularly in adolescents and young adults. Experimental research could also explore whether reducing TikTok usage or changing viewing habits (such as limiting evening use) can improve attention and behavioral regulation.

For educators and mental health professionals, these results highlight the need to raise awareness about the potential cognitive impacts of high social media consumption. In addition, app developers and platform regulators may consider implementing design changes or screen time reminders to help mitigate potential negative effects on attention and impulse control. In general, these findings underscore the importance of balancing digital media engagement with strategies that support sustained attention and cognitive well-being, especially among younger users.

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